

# Merkle-Damgård hash function

## Introduction

Imagine sending a file to your friend's device in the other side of the planet. The device received the file, but now there is one tiny issue. How does he know if the file has been tampered with? Usually, sending a file into the communication channel required encryption algorithms to encrypt the file so that it remains confidential. What about its integrity? That is the job of hash function. This report will explore more about the Merkle-Damgård hash function, test it against SHA-256, the most widely used hash function currently, with various attacks.

## Technical background

### What is a hash function? [2]

Hashing in the world of IT is a method that convert an input into a fixed size and unique output. It is used in many different ways such as searching for data or storing password. As mentioned in above, hash function is used for verifying the integrity of a file after transportation. Not only file, but any type of data. The fixed size and unique output is known as a hash value. It works as a fingerprint for the data and thus, it is used to check if the data have been changed before its arrival.

### Merkle-Damgård construction [1]

Merkle-Damgård construction is a hash framework used in MD5, SHA-1 and SHA-2. The construction consists of a compression function and an initialisation vector, which a fixed string. Furthermore, it is also a block-based hash function. In another word, the input will be divided into evenly sized blocks and therefore, the input has to be padded with bits if the size of the input is not divisible by the block size. Additionally, the final block is also encoded with the length of the original message. The algorithm starts with the initial block being compressed together with the initialisation vector, which gives the first hash value. The hash value is thereafter used to compress together with the next block, which gives the second hash value. It continues until the final hash value is achieved.

### Avalanche effect

asd

### Length extension attack

asd

# Test and results

asd

## Tasks

Task1:

- 1) To mask a value to 32 bits basically means to set a variable size to only 32 bits. This is necessary in python because a python variable can exceed 32 bits limit.
- 2) Bit shift manipulates the bits of a variable by moving the 1 and 0s further to the left or the right side. Bit rotations are similar to bit shift, but the only difference is when the bits move beyond the limit, those bits appear on the other end.  
Rotation is important because of how sensitive a hash is. If a bit is loss, the hash will be unusable.

Task2:

- 1) Compression function is used to compress the input into a fixed size output.
- 2) The combination of addition, XOR and rotation makes it hard to find a specific hash value with inputs other than the original input. This is the one-way property of a hash function

Task3:

- 1) Padding is necessary for dividing a given message to even-sized block.
- 2) Message length is necessary because otherwise it would be easy for attacker to perform length extension attack. (what is length extension attack?)
- 3) Preimage resistance is a property of a hash function that makes it difficult to compute the original input by using a hash function on a variable.  
Collision resistance is the other property that prevents two distinct inputs from giving the same hash value. A hash function has weak collision resistance if it only prevents collision with some specific inputs. A hash function with a strong collision resistance prevents collisions with every possible input.

Task4:

- 1) A well-designed hash function will give a completely different hash value even if the input has been changed slightly
- 2)

## Conclusion

The assignment is not completed

## **Bibliography**

- 1) Tamassia, Roberto & Goodrich, Michael. (2013). Introduction to computer security (1<sup>st</sup> edition). Introduction, chapter 3.5, Cryptography, chapter 3.1. Pearson international content.
- 2) Tamassia, Roberto & Goodrich, Michael. (2013). Introduction to computer security (1<sup>st</sup> edition). Introduction, chapter 3.5. Pearson international content.